#Header Section

Qarabağ University

Faculty of MATH and IT

Specialty: Computer Engineering

Course: Introduction to Python Programming

Laboratory Assignment № 1

Topic: Conditional Statements and Decision-Making in Python (`if`, `elif`, `else`)

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Instructor: Şamil Hümbətov

#Task Descriptions

Task 1: Body Mass Index (BMI) Calculator

Objective: Calculate the user's Body Mass Index (BMI) based on entered weight in kilograms and height in meters, and determine the corresponding health category.

Working Principle: The program uses the standard formula BMI = weight/(height*height). Based on the result, it uses `if-elif-else` conditional statements to classify the user as Underweight, Normal Weight, Overweight, or Obese.

Task 2: Chinese Zodiac Calculator

Objective: Determine the Chinese Zodiac animal sign based on the user's birth year.

Working Principle: The Chinese zodiac follows a 12-year cycle. The program calculates the remainder of the year divided by 12 (`year % 12`). Depending on the remainder (0 to 11), it displays the corresponding animal such as Monkey, Rooster, Dog, Pig, Rat, Ox, Tiger, Rabbit, Dragon, Snake, Horse, or Sheep.

Task 3: Birthday Guessing Game

Objective: Guess the user's birthday date of the month logically using 5 predefined number sets without asking the date directly.

Working Principle: The program displays 5 separate number sets and asks whether the birthday is included in each set (`1` for Yes, `0` for No). It sums up the first numbers of the sets where the user answered "Yes" to accurately calculate and display the birth date.

```
------------------------------
   PYTHON LAB 1 - MAIN MENU
==============================
1. BMI Calculator
2. Chinese Zodiac Calculator
3. Birthday Guessing Game
4. Exit
Select an option (1-4): 3

--- Birthday Guessing Game ---
set 1 : [4, 5, 6, 7, 12, 13, 14, 15, 20, 21, 22, 23, 28, 29, 30, 31]
Is your birthday included in this set?
Enter 1 for yes and 0 for no: 0
set 2 : [1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31]
Is your birthday included in this set?
Enter 1 for yes and 0 for no: 1
set 3 : [8, 9, 10, 11, 12, 13, 14, 15, 24, 25, 26, 27, 28, 29, 30, 31]
Is your birthday included in this set?
Enter 1 for yes and 0 for no: 1
set 4 : [16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31]
Is your birthday included in this set?
Enter 1 for yes and 0 for no: 0
set 5 : [2, 3, 6, 7, 10, 11, 14, 15, 18, 19, 22, 23, 26, 27, 30, 31]
Is your birthday included in this set?
Enter 1 for yes and 0 for no: 0
Your Birthday is: 9
```

```
                    print("your zodiac sign is tiger")
                elif a == 7:
                    print("your zodiac sign is rabbit")
                elif a == 8:
                    print("your zodiac sign is dragon")
                elif a == 9:
                    print("your zodiac sign is snake")
                elif a == 10:
                    print("your zodiac sign is horse")
                else:
                    print("your zodiac sign is goat")

            except ValueError:
                print("Invalid input! Please enter a valid integer year.")

        elif choice == '3':
            print("\n--- Birthday Guessing Game ---")
            try:
                setfirst = [4,5,6,7,12,13,14,15,20,21,22,23,28,29,30,31]
                setsecond = [1,3,5,7,9,11,13,15,17,19,21,23,25,27,29,31]
                setthird = [8,9,10,11,12,13,14,15,24,25,26,27,28,29,30,31]
                setfourth = [16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31]
                setfifth = [2,3,6,7,10,11,14,15,18,19,22,23,26,27,30,31]
                sets = [setfirst, setsecond, setthird, setfourth, setfifth]

                day = 0
                for i in range(5):
                    print("set", i + 1, ":", sets[i])
                    print("Is your birthday included in this set?")
                    answer = int(input("Enter 1 for yes and 0 for no: "))

                    if answer == 1:
                        day = day + sets[i][0]

                print("Your Birthday is:", day)

            except ValueError:
                print("Invalid input! Please enter 1 or 0.")

        elif choice == '4':
            print("Exiting program. Goodbye!")
            break
        else:
            print("Invalid choice! Please select a number between 1 and 4.")

if __name__ == "__main__":
    main()
```

*cbsdbhsdbchsdbchsdbc.py - C:/Users/user/Downloads/cbsdbhsdbchsdb

File Edit Format Run Options Window Help

```
year = int(input("Enter your born year:"))
a = year % 12
if a == 0:
    print("your zodiac sign isn monkey")
elif a == 1:
    print("your zodiac sign is rooster")
elif a ==2:
    print("your zodiac sign is dog")
elif a == 3:
    print("your zodiac sign is pig")
elif a == 4:
    print("your zodiac sign is rat")
elif a == 5:
    print("your zodiac sign is ox")
elif a == 6:
    print("your zodiac sign is tiger")
elif a == 7:
    print("your zodiac sign is rabbit")
elif a == 8:
    print("your zodiac sign is dragon")
elif a == 9:
    print("your zodiac sign is snake")
elif a == 10:
    print("your zodiac sign is horse")
else:
    print("your zodiac sign is goat")
```

*cbsdbhsdbchsdbchsdbc.py - C:/Users/user/Downloads/cbsdb

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```
kq = float(input("Enter your weight:"))
boy = float(input("Enter your height:"))
BMI = (kq) / (boy**2)
print(BMI)
if BMI < 18.5:
    print("You are underweight, thin")
elif BMI > 18.5 and BMI < 25:
    print("you are weight is normal")
elif BMI > 25 and BMI < 30:
    print("you are overweight")
else:
    print("You are obese, obesity")
```

```
set 1 : [4, 5, 6, 7, 12, 13, 14, 15, 20, 21, 22, 23, 28, 29, 30, 31]
is your birthday included in this set?
ENter 1 for yes and 0 for no:0
set 2 : [1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31]
is your birthday included in this set?
ENter 1 for yes and 0 for no:1
set 3 : [8, 9, 10, 11, 12, 13, 14, 15, 24, 25, 26, 27, 28, 29, 30, 31]
is your birthday included in this set?
ENter 1 for yes and 0 for no:1
set 4 : [16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31]
is your birthday included in this set?
ENter 1 for yes and 0 for no:0
set 5 : [2, 3, 6, 7, 10, 11, 14, 15, 18, 19, 22, 23, 26, 27, 30, 31]
is your birthday included in this set?
ENter 1 for yes and 0 for no:0
your birthday is : 9
```

```
= RESTART: C:/Users/user/Downloads/cbsdbhsdbchsdbc
Enter your weight:70
Enter your height:1.75
22.857142857142858
you are weight is normal
>>>
```

```
setfirst=[4,5,6,7,12,13,14,15,20,21,22,23,28,29,30,31]
setsecond = [1,3,5,7,9,11,13,15,17,19,21,23,25,27,29,31]
setthird = [8,9,10,11,12,13,14,15,24,25,26,27,28,29,30,31]
setfourth = [16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31]
setfifth = [2,3,6,7,10,11,14,15,18,19,22,23,26,27,30,31]
sets = [setfirst,setsecond,setthird,setfourth,setfifth]
day = 0
for i in range(5):
    print("set", i+ 1, ":", sets[i])
    print("is your birthday included in this set?")
    answer = int(input("ENter 1 for yes and 0 for no:"))
    if answer == 1:
        day = day + sets[i][0]
print("your birthday is :", day)
```

```
set 1 : [4, 5, 6, 7, 12, 13, 14, 15, 20, 21, 22, 23, 28, 29, 30, 31]
is your birthday included in this set?
ENter 1 for yes and 0 for no:0
set 2 : [1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31]
is your birthday included in this set?
ENter 1 for yes and 0 for no:1
set 3 : [8, 9, 10, 11, 12, 13, 14, 15, 24, 25, 26, 27, 28, 29, 30, 31]
is your birthday included in this set?
ENter 1 for yes and 0 for no:1
set 4 : [16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31]
is your birthday included in this set?
ENter 1 for yes and 0 for no:0
set 5 : [2, 3, 6, 7, 10, 11, 14, 15, 18, 19, 22, 23, 26, 27, 30, 31]
is your birthday included in this set?
ENter 1 for yes and 0 for no:0
your birthday is : 9
```

```
------------------------------
   PYTHON LAB 1 - MAIN MENU
==============================
1. BMI Calculator
2. Chinese Zodiac Calculator
3. Birthday Guessing Game
4. Exit
Select an option (1-4): 3

--- Birthday Guessing Game ---
set 1 : [4, 5, 6, 7, 12, 13, 14, 15, 20, 21, 22, 23, 28, 29, 30, 31]
Is your birthday included in this set?
Enter 1 for yes and 0 for no: 0
set 2 : [1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31]
Is your birthday included in this set?
Enter 1 for yes and 0 for no: 1
set 3 : [8, 9, 10, 11, 12, 13, 14, 15, 24, 25, 26, 27, 28, 29, 30, 31]
Is your birthday included in this set?
Enter 1 for yes and 0 for no: 1
set 4 : [16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31]
Is your birthday included in this set?
Enter 1 for yes and 0 for no: 0
set 5 : [2, 3, 6, 7, 10, 11, 14, 15, 18, 19, 22, 23, 26, 27, 30, 31]
Is your birthday included in this set?
Enter 1 for yes and 0 for no: 0
Your Birthday is: 9
```

```
==============================
   PYTHON LAB 1 - MAIN MENU
==============================
1. BMI Calculator
2. Chinese Zodiac Calculator
3. Birthday Guessing Game
4. Exit
Select an option (1-4): 1

--- BMI Calculator ---
Enter your weight: 70
Enter your height: 1.75
BMI: 22.857142857142858
Category: Normal
```

```
==============================
   PYTHON LAB 1 - MAIN MENU
==============================
1. BMI Calculator
2. Chinese Zodiac Calculator
3. Birthday Guessing Game
4. Exit
Select an option (1-4): 4
Exiting program. Goodbye!
```

```
                    print("your zodiac sign is tiger")
                elif a == 7:
                    print("your zodiac sign is rabbit")
                elif a == 8:
                    print("your zodiac sign is dragon")
                elif a == 9:
                    print("your zodiac sign is snake")
                elif a == 10:
                    print("your zodiac sign is horse")
                else:
                    print("your zodiac sign is goat")

            except ValueError:
                print("Invalid input! Please enter a valid integer year.")

        elif choice == '3':
            print("\n--- Birthday Guessing Game ---")
            try:
                setfirst = [4,5,6,7,12,13,14,15,20,21,22,23,28,29,30,31]
                setsecond = [1,3,5,7,9,11,13,15,17,19,21,23,25,27,29,31]
                setthird = [8,9,10,11,12,13,14,15,24,25,26,27,28,29,30,31]
                setfourth = [16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31]
                setfifth = [2,3,6,7,10,11,14,15,18,19,22,23,26,27,30,31]
                sets = [setfirst, setsecond, setthird, setfourth, setfifth]

                day = 0
                for i in range(5):
                    print("set", i + 1, ":", sets[i])
                    print("Is your birthday included in this set?")
                    answer = int(input("Enter 1 for yes and 0 for no: "))

                    if answer == 1:
                        day = day + sets[i][0]

                print("Your Birthday is:", day)

            except ValueError:
                print("Invalid input! Please enter 1 or 0.")

        elif choice == '4':
            print("Exiting program. Goodbye!")
            break
        else:
            print("Invalid choice! Please select a number between 1 and 4.")

if __name__ == "__main__":
    main()
```

```python
def main():
    while True:
        print("\n==============================")
        print("   PYTHON LAB 1 - MAIN MENU   ")
        print("==============================")
        print("1. BMI Calculator")
        print("2. Chinese Zodiac Calculator")
        print("3. Birthday Guessing Game")
        print("4. Exit")

        choice = input("Select an option (1-4): ")

        if choice == '1':
            print("\n--- BMI Calculator ---")
            try:
                kq = float(input("Enter your weight: "))
                boy = float(input("Enter your height: "))

                if kq <= 0 or boy <= 0:
                    print("Error: Weight and height must be greater than zero!")
                    continue

                BMI = (kq) / (boy**2)
                print(f"BMI: {BMI}")

                if BMI < 18.5:
                    print("You are underweight, thin")
                elif 18.5 <= BMI < 25:
                    print("you are weight is normal")
                elif 25 <= BMI < 30:
                    print("you are overweight")
                else:
                    print("You are obese, obesity")

            except ValueError:
                print("Invalid input! Please enter numeric values only.")

        elif choice == '2':
            print("\n--- Chinese Zodiac Calculator ---")
            try:
                year = int(input("Enter your born year: "))
                a = year % 12

                if a == 0:
                    print("your zodiac sign is monkey")
                elif a == 1:
                    print("your zodiac sign is rooster")
```

```
= RESTART: C:/Users/user/Downlc
Enter your born year:2009
your zodiac sign is ox
>>
```